PROPOSED BUDGET

Fungal and microbial colonisation of tropical educational building exteriors: molecular characterisation and campus air-quality context at UTS Sibu

Dr Armstrong Ighodalo Omoregie | BERI, School of Built Environment, UTS | UTS Matching Grant, Cycle 1/2027 | Year 1: Mar 2027 to Feb 2028, Year 2: Mar 2028 to Feb 2029

VOT	Category	Justification and detailed breakdown (Year 1 and Year 2)	Year 1 (RM)	Year 2 (RM)	Total (RM)
11000	**Salary and Wages**	**Y1 (RM 1,500):** Undergraduate research assistant allowance (RM 7.00/hour x 4 hours/day x approx. 18 days = RM 500.00/month x 3 active months) for sampling rounds R1 to R3 and monitoring node assembly. **Y2 (RM 1,500):** RA allowance (RM 500.00/month x 3 active months) for rounds R4 to R6, plate inoculation and cold-chain packing. Rate complies with Guideline Section 7(a); total within the RM 3,000 ceiling.	1,500.00	1,500.00	3,000.00
21000	**Travel and Transport**	**Y1 (RM 850):** Cold-chain transport of swabs, media and inoculated plates between UTS Sibu and SRDC-TROPI, Kota Samarahan (5 legs x RM 150.00 = RM 750.00); inbound delivery of sensor modules and field consumables (2 x RM 50.00 = RM 100.00). **Y2 (RM 2,150):** Cold-chain transport (4 legs x RM 150.00 = RM 600.00); delivery (RM 50.00); attendance at one national conference to present project findings (RM 1,500.00 covering return airfare RM 700.00, accommodation RM 250.00/night x 3 nights, ground transport RM 50.00). *(Note: Conference registration fee is claimed under VOT 29000. Local Sibu fieldwork travel excluded, as all four sampled buildings are on campus. Within RM 3,000 SET ceiling, Guideline Section 7(b).)*	850.00	2,150.00	3,000.00
24000	**Rental**	**Y1 (RM 1,200):** Rental of a reference-grade air quality analyser for co-location calibration of the monitoring nodes (RM 600.00/week x 2 weeks): one pre-deployment campaign and one post-installation verification. **Y2 (RM 0):** No further rental required. *(Note: Rental rather than purchase, as the instrument is required for two short campaigns only. Placed in VOT 24000 per Guideline Section 7(c).)*	1,200.00	0.00	1,200.00
26000	**Research Materials and Supplies**	**Y1 (RM 9,300):** Monitoring node components for four campus nodes plus spares (RM 8,040) comprising multi-parameter air-quality modules for CO2, PM2.5, PM10, temperature and humidity (RM 2,600); discrete temperature and humidity sensors (RM 200); metal oxide gas sensor sets (RM 1,120); microcontroller boards (RM 540); LoRa gateway and transmitter set (RM 900); IP65 enclosures and mounting (RM 880); power supply, solar panel, battery and cabling (RM 1,200); PCB fabrication and prototyping consumables (RM 600). Field sampling consumables (RM 1,260) comprising sterile sampling templates (RM 600), cold-chain coolers and gel packs (RM 360), and PPE for rounds R1 to R3 (RM 300). **Y2 (RM 2,500):** Additional cold-chain coolers (RM 360); PPE and field consumables for rounds R4 to R6 (RM 400); ethanol, sample bags, labels and markers (RM 540); culture consumable contingency (RM 1,200) for SDA and PDA media, sterile swabs and Petri dishes. *(Note: Node configuration is already validated in deployed service, removing design risk. Culture media and swabs are supplied by SRDC-TROPI as in-kind matched expenditure; the contingency covers control plates and repeat sampling only. Sulphur dioxide sensing excluded.)*	9,300.00	2,500.00	11,800.00
29000	**Professional Services**	**Y1 (RM 1,300):** Sanger sequencing of ITS amplicons, 32 reactions at RM 25.00 (RM 800.00) for isolates from rounds R1 to R3; SORAS application and Sarawak Biodiversity Council permit fees (RM 500.00). **Y2 (RM 1,700):** Sanger sequencing, 48 reactions at RM 25.00 (RM 1,200.00) for isolates from rounds R4 to R6; registration fee for one national conference (RM 500.00). *(Note: DNA extraction and PCR are performed by SRDC-TROPI as an in-kind matched contribution, so only the sequencing service is charged to this grant. Journal article processing charges excluded; non-APC journals targeted. Within RM 3,000 SET ceiling, Guideline Section 7(f).)*	1,300.00	1,700.00	3,000.00
35000	**Accessories and Small Appliances**	**Y1 (RM 3,000):** Outdoor-rated display panel for live campus air-quality readings at a high-traffic location (RM 2,200.00); weatherproof housing, mounting bracket and installation hardware (RM 800.00). **Y2 (RM 0):** No accessories required. *(Note: Cloud hosting and data plan excluded as utilities, provided instead as a UTS in-kind contribution. Assets remain UTS property. Total is 12% of grant, within the 40% ceiling, Guideline Section 7(g).)*	3,000.00	0.00	3,000.00
	TOTAL		**17,150.00**	**7,850.00**	**25,000.00**

VOT 23000 and VOT 28000 are not used. Virement may be exercised only once during the project and requires the approval of the Director, Centre for Research and Development. Supplier unit costs are planning figures pending current quotations, except the Sanger sequencing rate of RM25.00 per reaction, which is quoted.